

Sales Data Analysis & Business Intelligence Dashboard

Data Cleaning, EDA & Interactive Dashboard using Excel and Power BI

Presentation By

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 <https://github.com/salonitech-ai/sales-data-cleaning-project>



Project Objective

This project focuses on analyzing sales data to identify business trends, profitable products, and city-wise sales performance. The goal was to clean raw data, perform exploratory analysis, and build an interactive dashboard for business insights.



Our Objective

cleaned_sales_dataset.xlsx - Excel

Search

Draw Page Layout Formulas Data Review View Help

Calibri 11 A⁺ A⁻ B I U Font Merge & Center Alignment General Number Styles Cells Editing

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	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
City	Mumbai	2023-11-11	Phone	Accessories	13558.24	4.00	7724.71	Profit	1574.95%										
	Pune	2023-08-06	Laptop	Accessories	49460.12	5.00	6154.45	Profit											
	Mumbai	2023-12-15	Camera	Accessories	42789.26	1.00	13061.46	Profit											
	Delhi	2023-10-07	Camera	Electronics	47662.54	1.00	14892.65	Profit											
	Delhi	2023-08-13	Headphones	Electronics	38624.99	3.00	6970.81	Profit											
	Bangalore	2023-08-25	Phone	Electronics	16184.28	4.00	-410.99	loss											
	Mumbai	2023-10-21	Headphones	Accessories	36764.46	4.00	6904.11	Profit											
	Delhi	2023-05-31	Phone	Accessories	22808.10	1.00	6439.00	Profit											
	Pune	2023-02-11	Phone	Electronics	26566.79	2.00	6290.13	Profit											
	Delhi	2023-03-21	Phone	Accessories	35541.41	1.00	11068.01	Profit											
	Pune	2023-05-17	Tablet	Accessories	27997.00	2.00	-1706.51	loss											
	Chennai	2023-05-12	Phone	Electronics	48039.93	5.00	10866.16	Profit											
	Mumbai	2023-01-04	Phone	Electronics	27619.44	2.00	2418.38	Profit											
	Chennai	2023-09-03	Laptop	Accessories	5622.95	1.00	10960.44	Profit											
	Mumbai	2023-01-06	Phone	Electronics	3781.86	1.00	7529.26	Profit											
	Mumbai	2023-06-20	Phone	Electronics	9312.04	5.00	4763.75	Profit											
	Mumbai	2023-01-03	Camera	Accessories	34023.93	4.00	3371.38	Profit											
	Bangalore	2023-03-12	Headphones	Accessories	2741.45	1.00	10981.62	Profit											
	Bangalore	2023-10-07	Headphones	Accessories	8485.59	1.00	-1138.87	loss											
	Mumbai	2023-04-12	Camera	Electronics	32607.77	5.00	13975.30	Profit											
	Delhi	2023-02-23	Tablet	Electronics	32935.35	5.00	3496.69	Profit											
	Delhi	2023-03-29	Headphones	Electronics	28410.32	3.00	12619.50	Profit											
	Bangalore	2023-01-29	Phone	Electronics	36398.28	3.00	-3671.99	loss											
	Delhi	2023-11-28	Tablet	Accessories	34503.80	5.00	2874.45	Profit											
	Delhi	2023-12-01	Headphones	Electronics	23931.49	5.00	4863.54	Profit											
	Mumbai	2023-11-26	Camera	Electronics	37148.74	5.00	3841.05	Profit											
	Bangalore	2023-07-31	Headphones	Accessories	43210.37	5.00	-4204.36	loss											
	Pune	2023-08-10	Camera	Electronics	21055.36	2.00	10788.47	Profit											

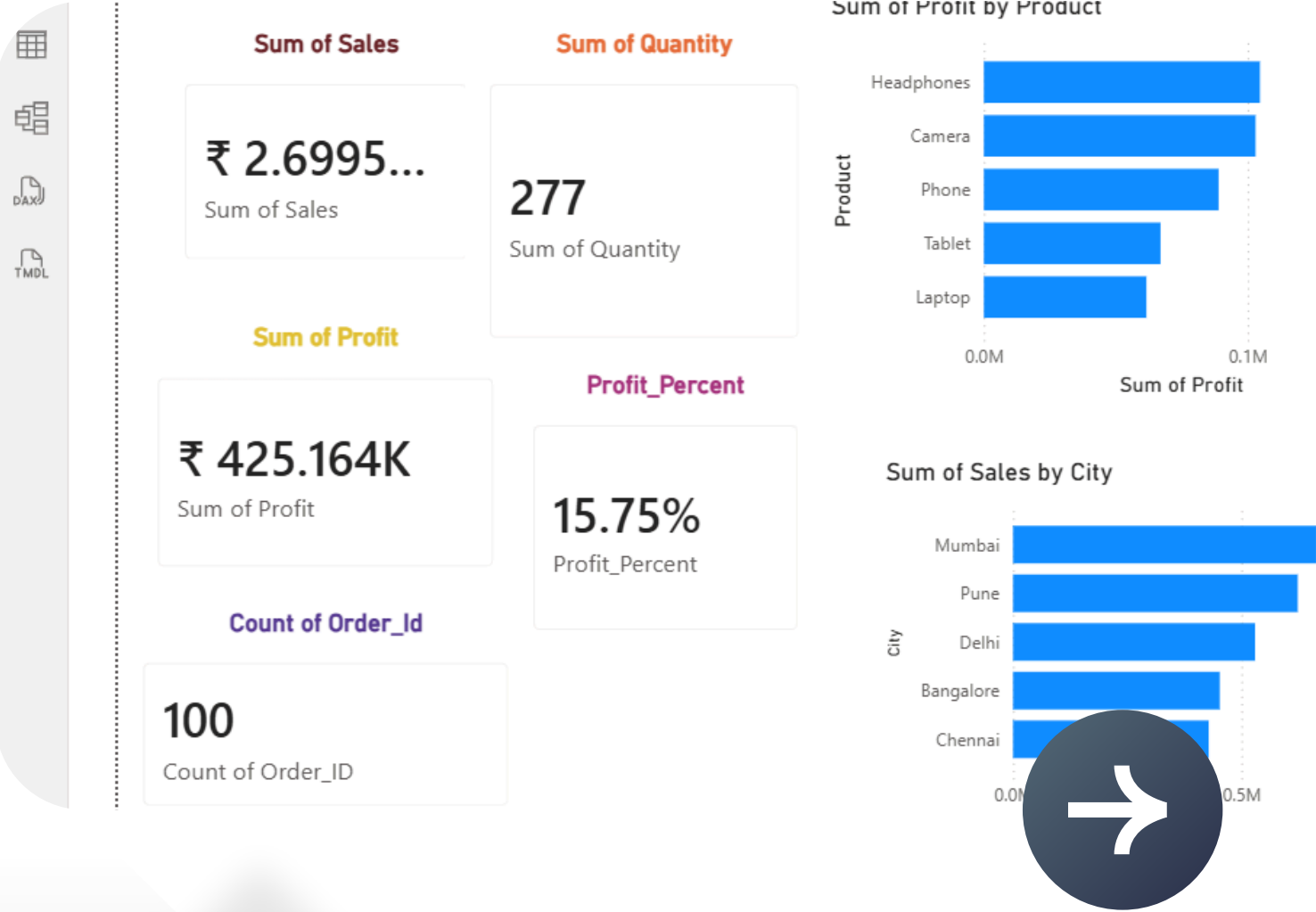
ed data set EDA summary Sales By City Profit By Product Profit vs Loss Insights KPI

Investigate

Windows Search

- Removed duplicate records
- Handled missing values
- Corrected data types
- Standardized categorical values
- Created Profit_Type column
- Prepared clean dataset for analysis

Key Performance Indicators



KPI	Value
Total Sales	2.70M
Total Profit	425.16K
Total Orders	100
Total Quantity	277
Profit Margin	15.75%



Interactive Dashboard



The dashboard provides interactive analysis of sales, profit, quantity, products, and city-wise performance.



Key Insights

- **Mumbai recorded the highest sales.**
- **Chennai showed the lowest sales performance.**
- **Headphones generated the highest profit.**
- **Laptop category showed comparatively lower profitability.**
- **71% of orders were profitable.**
- **Electronics category contributed significantly to revenue.**



Statistical Validation

Hypothesis:

Headphones generate higher profit than laptops.

Result:

Analysis of sales data indicates that headphones consistently produced higher profits compared to laptops.

Business Conclusion:

The company should focus more on high-profit products like headphones to improve overall profitability.



Recommendations

- Increase marketing campaigns in Mumbai.
- Improve sales strategy in Chennai.
- Focus inventory on profitable products.
- Reduce dependency on low-performing products.
- Use dashboard insights for future decision-making.



Conclusion

This project successfully transformed raw sales data into actionable business insights using Excel and Power BI. The dashboard enables better business decisions through interactive visualization and KPI tracking.